

ASSIGNMENT 2
DAIRY HERD REPRODUCTION
ANS 235
2/10/26
DUE: 2/28/26

You are an owner of a 1200-cow dairy and are considering changing the reproductive program on your dairy to the proposed situation below. Your answers will need to be uploaded to D2L. The easiest way to complete this assignment is to build an excel spreadsheet. However, that is not a requirement to complete!

When you upload your file or files please use your last name as the file name.

Current Situation

VWP = 60 Days
Heat Detection (Service Rate) = 38%
Conception Rate = 33%
Cull cows after 11 estrous cycles.

Proposed Situation

VWP = 80 Days
Heat Detection Rate (S. Rate) = 72%
Conception Rate = 49%
Cull cows after 11 estrous cycles

For each scenario, answer the following questions. **PLEASE SHOW YOUR WORK FOR FULL CREDIT!**

1. What is the 21-day pregnancy rate for each situation?

Current situation: $PR = CR \times HDR = 0.33 \times 0.38 = 0.1254 = 12.54\%$

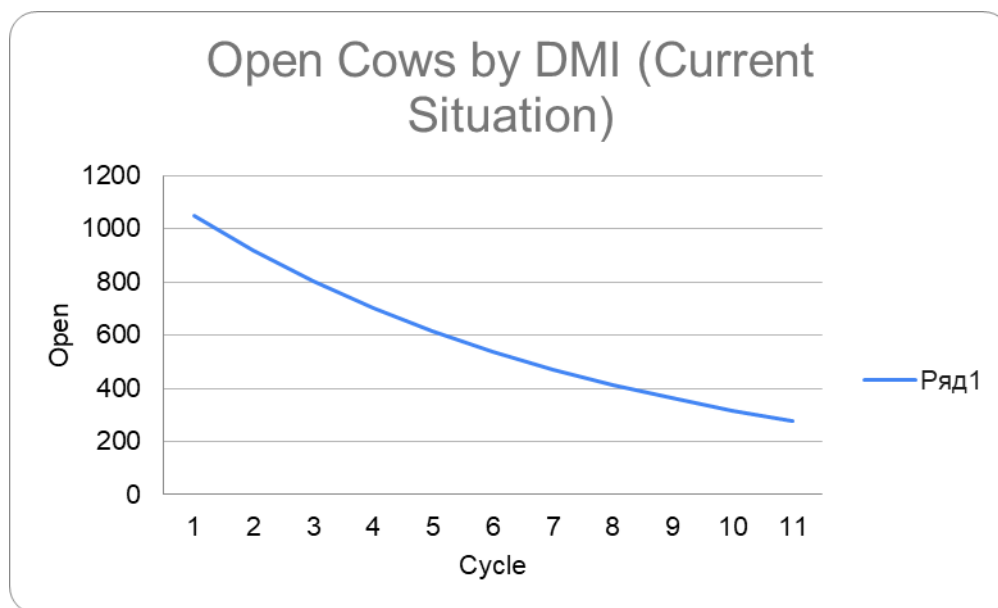
Proposed Situation: $PR = 0.49 \times 0.72 = 0.3528 = 35.28\%$

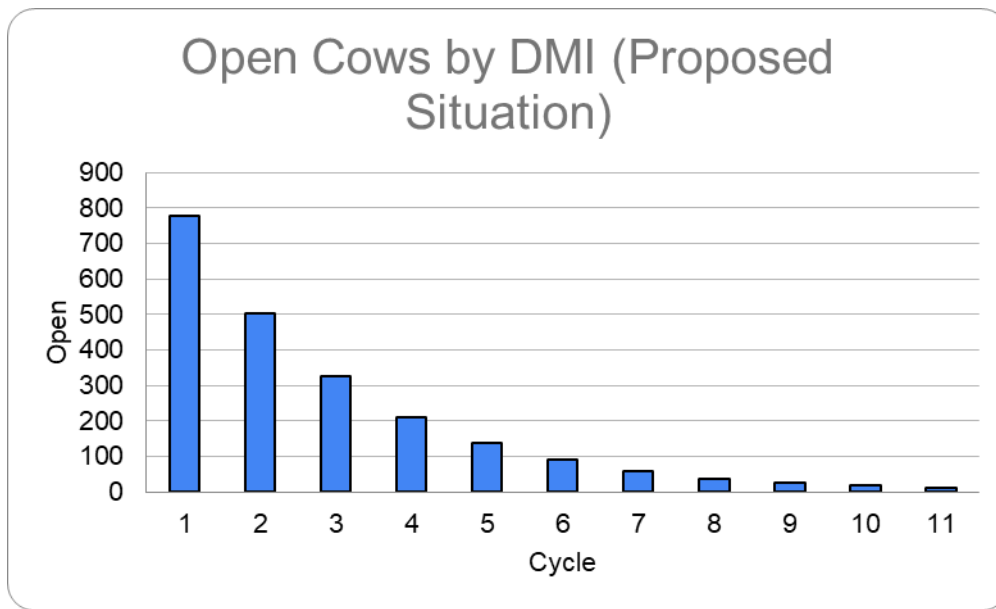
2. What is the distribution of cows in heat (or serviced), cows pregnant, cows open and total cows pregnant after each estrous cycle for each situation above. This should be provided in a table format.

Cycle Current	Start Open	HDR (38%)	CR (33% of Serviced)	Open	Total Pregnant (12.54%)	DMI
1	1200	456	396	1050	150	80
2	1050	399	346	919	131	100
3	919	349	303	804	115	120
4	804	305	265	704	100	140
5	704	267	232	616	88	160
6	616	234	203	539	77	180
7	539	204	177	472	67	200
8	472	179	155	413	59	220
9	413	156	136	362	51	240
10	362	137	119	317	45	260
11	317	120	104	278	39	280

Cycle Proposed	Start Open	HDR (72%)	CR (49% of Serviced)	Open	Total Pregnant (35,28%)
1	1200	864	588	777	423
2	777	559	256	503	274
3	503	362	165	326	177
4	326	234	107	211	115
5	211	151	69	137	74
6	137	98	45	89	48
7	89	64	29	58	31
8	58	41	19	38	20
9	38	27	12	25	13
10	25	18	8	17	8
11	17	12	5	12	5

3. Provide a graph of open cows by Day in Milk (DIM) for each situation.





4. How many total cows are pregnant at 140 DIM for each situation?

Current situation: 496 cows

Proposed: 989 cows

5. How many cows are open after 11 estrous cycles for each situation?

Remaining open cows in *current situation* after the 11th cycle - **278**;

Remaining open cows in *proposed situation* after the 11th cycle - **12**.

6. What is the average days open (ADO) for the current and proposed situation?

$$VWP + \left(\frac{1}{PR} - 0.5 \right) * 20$$

Current situation:

$$60 + \left(\frac{1}{0.1254} - 0.5 \right) * 20 \approx 210 \text{ days}$$

Proposed situation:

$$80 + \left(\frac{1}{0.3528} - 0.5 \right) * 20 \approx 127 \text{ days}$$

7. Discuss **in detail** the pros and cons of each situation.

One of the disadvantages of current situation, the number of open cows at

the end of the cycle is high. Moreover, ADO is much larger compare to the benchmark. As a result, we won't get the desired number of calves, and also we will see a decrease in the dairy production.

One of the advantages of current situation compare to the proposed situation is in the effort we put to detect the heat (HDR). A High HDR requires higher workload and more expenses.

However, Proposed Situation offers way more benefits than its drawbacks: less open cows at the cycle end results a large amount of calves. 99% of 1200 cows were conceived.

DUE February 28, 2026 by Noon